

From Double to Triple

In Sections 15.1–15.5 we integrated $f(x, y)$ over a 2D region in $\mathbb{R}^2$. Now we go one dimension higher: integrating $f(x, y, z)$ over a 3D solid in $\mathbb{R}^3$.

Recall the progression:

- **Calc I:** $\int_a^b f(x) dx$ — integrate over an interval (1D)
- **Calc III (Ch 15.1–15.5):** $\iint_D f(x, y) dA$ — integrate over a region (2D)
- **Now:** $\iiint_E f(x, y, z) dV$ — integrate over a solid (3D)

The construction is the same: partition the solid E into tiny boxes, multiply each f -value by the volume of the box $\Delta V = \Delta x \Delta y \Delta z$, add them all up, and take the limit:

$$\iiint_E f(x, y, z) dV = \lim_{l, m, n \rightarrow \infty} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \Delta V$$

Same idea, one more dimension. Chop, multiply, add, take the limit.

Interpretation: Density and Mass

What does $\iiint_E f(x, y, z) dV$ mean physically?

If $f(x, y, z) = 1$ for all points, then:

$$\iiint_E 1 dV = \text{volume of } E$$

This is the 3D version of "area = $\iint_D 1 dA$."

The most common physical interpretation: if $f(x, y, z) = \rho(x, y, z)$ is a **density function** (mass per unit volume, like g/cm^3 or kg/m^3), then

$$\iiint_E \rho(x, y, z) dV = \text{total mass of } E$$

More generally, if f gives "amount per unit volume," then $\iiint_E f dV$ gives the total amount in E :

- Charge density (C/m^3) $\rightarrow$ total charge
- Bacteria concentration (cells/cm^3) $\rightarrow$ total bacteria count
- Probability density $\rightarrow$ total probability

Triple integrals accumulate a quantity spread throughout a 3D solid — density times volume gives mass.

Quick Check: Interpret a Triple Integral

Suppose $\rho(x, y, z) = 3z$ gives the density (in g/cm^3) of a solid E .

What does $\iiint_E 3z dV$ compute? What are the units?

Think about what density $\times$ volume gives you. What role does z play in the density?

Setting Up the Limits of Integration

Just like double integrals, we evaluate triple integrals as **iterated integrals**. In the order $dz\,dy\,dx$:

$$\iiint_E f(x, y, z) \, dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y)}^{u_2(x,y)} f(x, y, z) \, dz \, dy \, dx$$

How to find the limits (working from outside in):

1. **Outer integral (dx):** What range of x -values does the solid span? These are constants.
2. **Middle integral (dy):** For a fixed x , what range of y -values? These may depend on x .
3. **Inner integral (dz):** For fixed x and y , what range of z -values? These may depend on both x and y .

Alternatively (working from inside out):

1. Identify the **top** and **bottom** surfaces $\rightarrow$ limits on z
2. **Project** the solid onto the xy -plane $\rightarrow$ a 2D region
3. Set up the double integral over that 2D region as in Sections 15.2–15.3

The hardest part of triple integrals is **finding the limits**. The actual integration is just three rounds of Calc I antiderivatives.

Example 1: Triple Integral Over a Tetrahedron

Example 1

Evaluate $\iiint_E 2xyz \, dV$, where E is the solid in the first octant bounded by the plane $3x + 2y + z = 1$.

Describe the solid: E is a **tetrahedron** (triangular pyramid) with vertices at $(0, 0, 0)$, $(\frac{1}{3}, 0, 0)$, $(0, \frac{1}{2}, 0)$, and $(0, 0, 1)$.

Find the limits for $dz\,dy\,dx$. Start with: for fixed x, y , z goes from 0 up to $z = 1 - 3x - 2y$.

Quick Check: Describe the Solid

Describe the solid E whose volume is given by:

$$\int_0^1 \int_0^{1-x} \int_0^{2-2x-2y} dz \, dy \, dx$$

Read the limits from outside in. What shape has these boundary surfaces?

Six Orders of Integration

A triple integral can be written in **six** different orders:

$$dz \, dy \, dx \quad dz \, dx \, dy \quad dy \, dz \, dx \quad dy \, dx \, dz \quad dx \, dz \, dy \quad dx \, dy \, dz$$

All six give the same answer (assuming the function is continuous), but some orders may be **much easier** to evaluate than others!

Strategy for choosing the best order:

- Look for limits that are **constants** — those variables should go on the outside
- Look for an integrand that simplifies when you integrate a particular variable first
- Sketch the solid and look for natural "top/bottom," "left/right," or "front/back" descriptions

For the tetrahedron below $3x + 2y + z = 1$ (Example 1), here's the $dy \, dz \, dx$ version:

Fix x and z : y goes from 0 to $\frac{1-3x-z}{2}$. **Fix x :** z goes from 0 to $1 - 3x$. **Then:** x goes from 0 to $\frac{1}{3}$.

$$\int_0^{1/3} \int_0^{1-3x} \int_0^{(1-3x-z)/2} 2xyz \, dy \, dz \, dx$$

Switching the order of integration changes the **limits**, not the **integrand**. Always re-derive the limits from scratch by thinking about the geometry.

Example 2: Another Tetrahedron

Example 2

Evaluate $\iiint_E y \, dV$, where E is bounded by the coordinate planes $x = 0$, $y = 0$, $z = 0$, and the plane $2x + 2y + z = 4$.

Set up the integral in order $dz \, dy \, dx$. What is the top surface? What is the projection onto the xy -plane?

Example 3: Volume Using a Triple Integral

Example 3

Find the volume of the solid bounded by the parabolic cylinder $y = x^2$, and the planes $z = 0$, $z = 4$, and $y = 9$.

Describe the solid: The solid sits between $z = 0$ and $z = 4$ (constant height), behind $y = 9$, and in front of $y = x^2$ (a parabolic wall).

Set up $\iiint_E 1 \, dV$ in order $dz \, dy \, dx$. Where does $y = x^2$ intersect $y = 9$?

Homework

Section 15.6: 1, 3, 5, 6, 7, 9, 11, 13, 15, 17, 19, 25, 31, 33